

Pre-Algebra Test I

As always, write a title, name, date, and course. Number each problem, circle the number, and show work where necessary. Draw a rectangle around your answers.

Simplify the following expressions.

① a) $4 - 8$

b) $-2 + 3$

c) $(-5) + (-6)$

d) $-8 + 6$

② a) $-7 + (-3)$

b) $-5 - 1$

c) $10 - (-2)$

d) $-4 - (-6)$

③ a) $4(-8)$

b) $-6(-2)$

c) $(-10)(-3)$

d) $(9)(-1)$

④ a) $-4(11)$

b) $-7(4)$

c) $(-3)(9)$

d) $(-4)(5)$

⑤ a) $\frac{-30}{5}$

b) $\frac{28}{-4}$

c) $\frac{-24}{-3}$

d) $-\frac{90}{10}$

⑥ a) $\frac{-14}{35}$

b) $\frac{16}{-40}$

c) $\frac{-5}{45}$

d) $-\frac{37}{36}$

⑦ a) $\frac{50}{81} - \frac{8}{9}$

b) $-\frac{4}{7} + \frac{11}{21}$

c) $-\frac{3}{4} + (-\frac{1}{5})$

d) $-\frac{2}{3} - \frac{1}{5}$

⑧ a) $\frac{1}{8} - (-\frac{1}{2})$

b) $-\frac{3}{8} - (-\frac{1}{4})$

c) $-\frac{3}{10}(\frac{2}{3})$

d) $-\frac{5}{6}(-\frac{2}{3})$

⑨ a) $\frac{2}{5} \div (-\frac{3}{8})$

b) $-\frac{7}{10} \div (-\frac{3}{4})$

c) $-2 + (-1) + (-7)$

d) $-3(-1)(-2)$

⑩ a) $-5 + 6 - 2$

b) $1 - 5 - 7$

c) $-10 - 2 - 8$

d) $7 + (-2) + 8 + (-10)$

⑪ a) $-3(-3)(-3)$

b) $4(2)(-1)$

c) $-\frac{5}{24} - \frac{1}{8} + \frac{7}{12}$

d) $-\frac{2}{9} + \frac{1}{6} - \frac{5}{18}$

$$\textcircled{12} \text{ a) } \frac{3}{4}(-2)(-\frac{1}{3})$$

$$\text{b) } -\frac{7}{10} - \frac{1}{2} - \frac{1}{5} + 4$$

$$\text{c) } -0.07(-1.9)$$

$$\text{d) } 3.6 - 8.11$$

$$\textcircled{13} \text{ a) } -4x + 7x$$

$$\text{b) } 5y - 10y$$

$$\text{c) } -11z - 12z$$

$$\text{d) } -8x + 5x$$

$$\textcircled{14} \text{ a) } 6x - (-x)$$

$$\text{b) } -3y - (-4y)$$

$$\text{c) } 3yz(-4y^2z^3)$$

$$\text{d) } \frac{88x^9}{11x^2}$$

$$\textcircled{15} \text{ a) } -2x^8y^6z(-3x^3yz)$$

$$\text{b) } 8x - 6x - 11x$$

$$\textcircled{16} \text{ a) } -9x^4y^6(-4xy^9z^7)$$

$$\text{b) } \frac{35a^3bc^2}{7a^2bc}$$

$$\textcircled{17} \text{ a) } \frac{-18a^5b^6c^7}{-24a^6b^7c}$$

$$\text{b) } -5x^9y^4(2x^4y^3z^4)$$

$$\textcircled{18} \text{ a) } \frac{-4x^2y^2z^3}{-28x^5y^5z}$$

$$\text{b) } \frac{3a^7b^5}{15a^3b^3}$$

$$\textcircled{19} \text{ a) } \frac{y}{2} - \frac{y}{4}$$

$$\text{b) } -\frac{3x}{7} + \frac{2x}{21}$$

$$\textcircled{20} \text{ a) } -10x + 6x - 8x$$

$$\text{b) } -3z - z - 8z$$

$$\textcircled{21} \frac{6x^3y^4}{11} \left(-\frac{2x^3z^2}{5y} \right)$$

$$(22) - \frac{4x^2y^5}{7z^2} \div \left(\frac{5y^8}{9x^5z} \right)$$

$$(23) - \frac{5x}{12} - \left(-\frac{x}{4} \right) + \frac{5x}{6}$$

$$(24) - \frac{8a^7}{3b^3c^2} \div (-5abc)$$

$$(25) (-10x^5y) \left(-\frac{4z^3y^{10}}{5x^4} \right)$$

Pre-Algebra Test I Answers

① a) -4 b) 1 c) -11 d) -2

② a) -10 b) -6 c) 12 d) 2

③ a) -32 b) 12 c) 30 d) -9

④ a) -44 b) -28 c) -27 d) -20

⑤ a) -6 b) -7 c) 8 d) -9

⑥ a) $-2/5$ b) $-2/5$ c) $-1/9$ d) $-3/4$

⑦ a) $-22/81$ b) $-1/21$ c) $-19/20$ d) $-13/15$

⑧ a) $5/8$ b) $-1/8$ c) $-1/5$ d) $5/9$

⑨ a) $-16/15$ b) $14/15$ c) -10 d) -6

⑩ a) -1 b) -11 c) -20 d) 3

⑪ a) -27 b) -8 c) $1/4$ d) $-1/3$

⑫ a) $1/2$ b) $13/5$ c) 0.133 d) -4.51

⑬ a) $3x$ b) $-5y$ c) $-23z$ d) $-3x$

⑭ a) $7x$ b) y c) $-12y^3z^4$ d) $8x^7$

⑮ a) $6x^{11}y^7z^3$ b) $-9x$

⑯ a) $36x^5y^{15}z^7$ b) $5ac$

⑰ a) $\frac{3c^6}{4ab}$ b) $-10x^{13}y^7z^4$

⑱ a) $\frac{z^2}{7x^3y^3}$ b) $\frac{a^4b^2}{5}$

⑲ a) $\frac{y}{4}$ b) $-\frac{x}{3}$

⑳ a) $-12x$ b) $-12z$

㉑ $-\frac{12x^6y^3z^2}{55}$

㉒ $-\frac{36x^7}{35y^3z}$

㉓ $\frac{2x}{3}$

㉔ $\frac{8a^6}{15b^4c^3}$

㉕ $8xy^{11}z^3$